

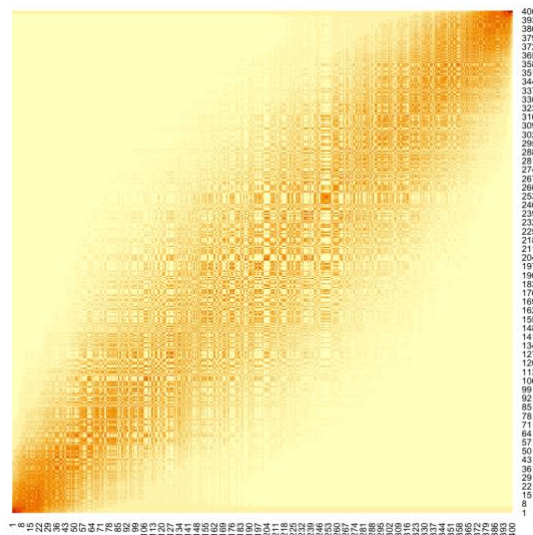
Spatial sample thinning

The idea behind my approach is based on the assumption that species distributions in nature follow a pattern similar to a Gaussian Process. In simple terms, it implies that each species tends to form one or more areas of high concentration—often referred to as “hotspots”—where individuals are densely aggregated. As one moves farther from these central points, the density of individuals gradually declines in a smooth, continuous manner expressed in the following form:

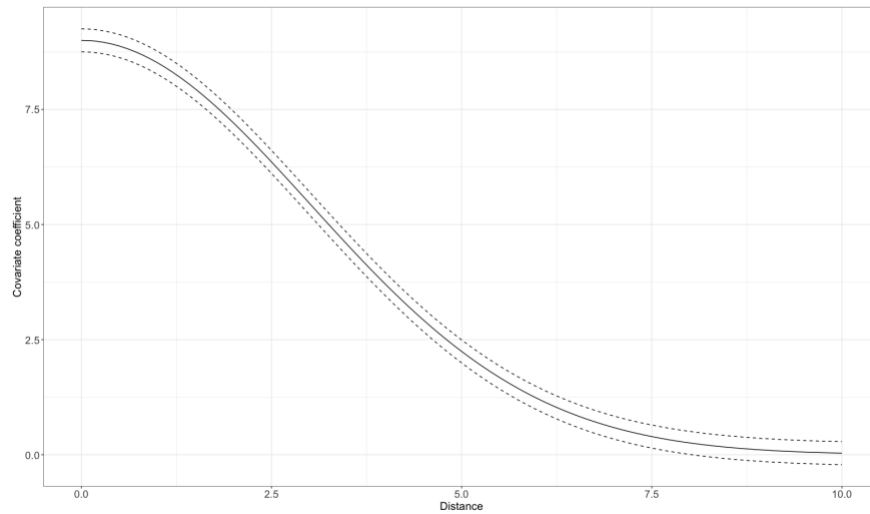
$$Y_i \sim MVN(\mu_i, k(x, x'))$$
$$k(x, x') = \alpha^2 * \exp \left(-\frac{1}{2\rho^2} * \|x - x'\|^2 \right) + \sigma^2$$

Where Y is the observed abundance or density of the fish species at location i with a mean abundance μ and covariate matrix k of coordinate x and x' . The covariate matrix is the most important part of the Gaussian Process (GP) as it determines the spatial autocorrelation of the species distribution where α is the amplitude of the x - x' correlation, ρ is the length-scale parameter meaning the correlation (in x - x' units) where the signal fades out, and σ is the noise in nature. In many GP models μ is typically set to 0 hence α^2 becomes the mean abundance.

Creating a covariate matrix for a GP process with spatial coordinates as the only covariates looks something like this figure where the diagonal matrix indicates the correlation coefficient between any given sampling points.



The link function of the $k(x-x')$ 2 can be summarized in this graph where the correlation coefficient will decay/fade out as the sampling point will move further away in distance.



In this link-function, 3 parameters are influencing the form of this relationship, α and ρ , while σ is the uncertainty around the line. Knowing or estimating these 2 parameters we can understand the species abundances and aggregation dynamics which can thereafter inform us on the sampling design. Given that we use eDNA as the measurement of species abundances and due to the environmental attributes of eDNA, such as transport, it will add noise to this function.